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PROJECT 2

BUS TICKET BOOKING SYSTEM

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1.0 Overview of the Project

The Bus Ticket Booking System is a web-based information system developed to improve and modernize the process of buying and managing bus tickets. The system is designed to provide customers with a convenient platform where they can view available bus schedules, check seat availability, select preferred seats, and complete ticket bookings online. Through the implementation of digital technology, the system aims to replace traditional manual booking practices with a more efficient, exact, and user-friendly solution.

In the transportation industry, buses stay one of the most widely used modes of public transport for both short-distance and long-distance travel. Every day, thousands of passengers rely on bus services for commuting to work, returning to their hometowns, attending educational institutions, and travelling for leisure purposes. As the demand for public transportation continues to increase, bus operators are expected to provide faster, more reliable, and more accessible services to meet customer expectations.

Despite technological advancements in many sectors, some bus ticket booking processes still depend on manual operations or outdated systems. Customers are often required to visit ticket counters physically to obtain information regarding schedules, seat availability, and ticket reservations. This process consumes time and effort, especially during peak travel periods when large numbers of customers try to secure tickets simultaneously.

The proposed Bus Ticket Booking System seeks to address these limitations by introducing an integrated online platform that allows customers to perform booking activities remotely. The system provides real-time access to bus schedules and seat availability, ensuring that users receive accurate and updated information before making travel decisions. By eliminating the dependency on manual processes, the system enhances convenience, reduces operational inefficiencies, and improves the overall customer experience.

In addition to benefiting customers, the system also provides significant advantages for bus operators and administrators. Through centralized data management, operators can efficiently monitor bookings, manage schedules, update seat allocations, and generate reports when required. All booking information is stored in a structured database, making data retrieval and management more systematic and reliable.

Overall, the Bus Ticket Booking System represents a digital transformation initiative within the transportation industry. By using information technology, the system aims to improve service quality, streamline business operations, reduce administrative workload, and support the growing need for efficient online transportation services.

2.0 Problem Statement

The current bus ticket booking process faces several operational and customer service challenges that affect the efficiency of the overall transportation service. These challenges arise primarily due to the continued reliance on manual procedures and the lack of a centralized digital platform for managing bookings and travel information.

One of the significant issues is the requirement for customers to be physically present at ticket counters to purchase bus tickets. This traditional approach is both time-consuming and inconvenient, particularly for customers who live far from bus terminals or have limited time availability. During weekends, public holidays, festive seasons, and school breaks, ticket counters often experience heavy congestion, resulting in long waiting queues and increased customer dissatisfaction. The need to travel to the counter solely for ticket reservations creates unnecessary inconvenience and reduces service accessibility.

Another significant problem is the absence of real-time information regarding bus schedules and seat availability. Customers often depend on information provided by staff members or notice boards, which may not always reflect the most recent updates. As a result, customers may encounter situations where seats are no longer available despite receiving earlier information indicating otherwise. This lack of transparency can create confusion, inconvenience, and frustration among passengers.

Furthermore, manual record-keeping practices increase the likelihood of human errors. Employees may accidentally enter incorrect customer information, misplace booking records, or process duplicate reservations. Such errors can lead to booking conflicts, customer complaints, and financial losses for the organization. Since manual processes rely heavily on human intervention, maintaining data accuracy becomes increasingly difficult as the number of bookings grows.

The current system also presents challenges for bus operators in managing operational data. Tracking ticket sales, monitoring seat occupancy, updating schedules, and generating reports require substantial administrative effort. Without an automated system, employees must manually compile information from various sources, which consumes time and reduces productivity. The lack of centralized data storage further complicates the process of retrieving historical booking information when needed.

Additionally, the existing booking process offers limited flexibility for customers. Modern consumers increasingly prefer digital services that allow transactions to be completed anytime and anywhere using internet-enabled devices. The inability to provide online booking services may place bus operators at a competitive disadvantage compared to transportation providers that have already adopted digital platforms.

Therefore, there is a strong need for an integrated and computerized Bus Ticket Booking System that can eliminate inefficiencies, reduce operational errors, improve data management, and provide a more convenient booking experience for customers while supporting the operational needs of bus service providers.

3.0 Proposed Solutions

To overcome the challenges found in the current booking process, a Bus Ticket Booking System is proposed as a comprehensive web-based solution. The system is designed to automate and streamline ticket reservation activities while providing a more convenient and efficient experience for both customers and bus operators.

The proposed system enables customers to access booking services through an online platform using devices such as smartphones, tablets, laptops, or desktop computers. Customers will be able to register accounts, browse available bus schedules, view seat availability in real time, select preferred seats, and complete ticket bookings without needing to visit a physical ticket counter. This functionality significantly reduces waiting times and provides greater flexibility for users to make bookings at their convenience.

One of the key features of the proposed solution is the implementation of a centralized database that stores and manages all booking-related information. Through this centralized approach, seat availability and bus schedules can be updated instantly whenever a booking is made or modified. Customers will therefore receive accurate and up-to-date information, reducing the possibility of booking conflicts and misunderstandings.

The system will also provide administrative functions for bus operators. Administrators will be able to create and manage bus schedules, update route information, monitor bookings, manage customer records, and generate operational reports. These functions help reduce the workload associated with manual record management while improving overall organizational efficiency.

In addition, the proposed system incorporates validation and verification mechanisms to improve data accuracy and prevent common errors. For example, the system can automatically prevent double bookings by updating seat availability immediately after a reservation is confirmed. Input validation can also reduce mistakes caused by incorrect data entry, ensuring that customer information remains accurate and complete.

The implementation of the Bus Ticket Booking System is expected to deliver numerous benefits. Customers will enjoy faster and more convenient booking services, improved access to travel information, and greater flexibility in managing their reservations. Meanwhile, bus operators will benefit from improved operational efficiency, reduced administrative costs, enhanced data management, and better decision-making through access to accurate booking records and reports.

Ultimately, the proposed solution aligns with current technological trends and supports the growing demand for digital transportation services. By replacing manual processes with an integrated online platform, the Bus Ticket Booking System will improve service quality, enhance customer satisfaction, and contribute to the long-term growth and sustainability of bus transportation operations.

4.0 Information Gathering

4.1 Method Used

4.1.1 Interview

An interview was conducted with potential users to gain a deeper understanding of the current bus ticket booking process and the challenges they face. Through direct communication, detailed information was obtained such as user's experience with bus ticket booking process, crucial information in a bus ticket booking system, and problems that users encounter when using the booking system. The interview provided valuable insights into users' expectations and preferences for the proposed system.

4.1.2 Survey

A questionnaire was created using Google Forms and distributed to potential users. The survey consisted of multiple-choice, checkbox, and Likert-scale questions designed to collect data on users' experiences with existing booking methods, the problem they encounter, and the features they would like to see in an online bus ticket booking system.

This method was chosen because it allows data to be collected from a larger number of respondents within a short period of time. The respondents within a short period of time. The responses obtained were analysed to identify user requirements and support the development of the proposed system.

By combining interviews and questionnaires, both detailed qualitative information and broader quantitative data were collected, resulting in a more comprehensive understanding of user needs and system requirements.

4.2 Summary from Method

Two methods were carried out to gather sufficient information regarding the feasibility of this project. These methods include interviewing and conducting an online questionnaire.

4.2.1 Interview

Based on the answers of the interviewees, we can surmise that online booking is a more popular choice as well as being more convenient to a regular bus user. Although Student 1 lamented having a confusing time during their first time trying to book a bus ticket, the experience is easy and smooth as confirmed by Student 2 once he got used to the process.

Both students state that the most important information required when booking a bus ticket is the date and time of the trip. It can be derived from this information that it is because the bus trip must be able to fit into the students' existing schedule instead of being built around on. This is important because a student's hectic day includes classes, assignments and occasionally, mandatory quizzes and tests. As such, the bus trip must be planned beforehand to avoid clashing with any of these affairs.

Student 1 and Student 2 reported different difficulties with booking online. Student 1 states that he has experienced double booking due to a lack of real-time information on the booking platform. As such, he was allowed to book an already booked seat. Student 2 however complained of a messy user interface that is hard to use as a customer. She wishes for a better structured user interface as future improvement.

From the interview, we have identified several pain points and important features. These features include a bus schedule and pre-booking bus trips while pain points are booking platforms' messy user interface and lack of real-time information. We plan to address these pain points in our project by introducing real-time seat information and a clean, easy-to-navigate user interface (UI).

4.2.2 Survey

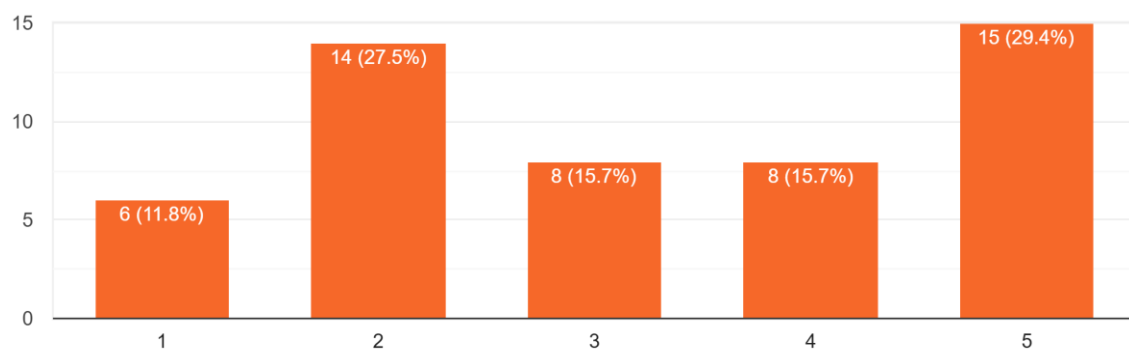
A total of 51 respondents participated in the questionnaire survey conducted through Google Form. The demographic information is collected first in Section A of the survey. In regards with the respondents' age group, 62.7% are in the age range of 18-25, 13.7% respectively for both age ranges 26-35 and those above 35, while the rest are aged below 18 which is 9.8%. Gender wise, 41.2% are female and the rest are male. For occupation, 68.6% are students, 19.6% are employees, 9.8% are business owners and the rest is retired.

Next, Section B investigates the respondent's experience with bus transportation. The survey results showed that most respondents have experience using bus transportation and have encountered issues with the current ticket booking process. Below attached the result charts from the survey.

Question 1:

How often do you use bus transportation per month?

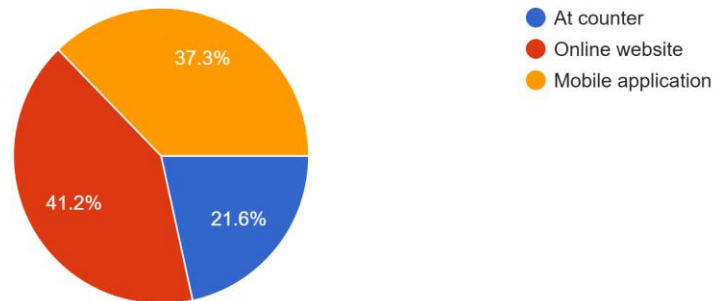
51 responses



Question 2:

How do you usually purchase bus ticket?

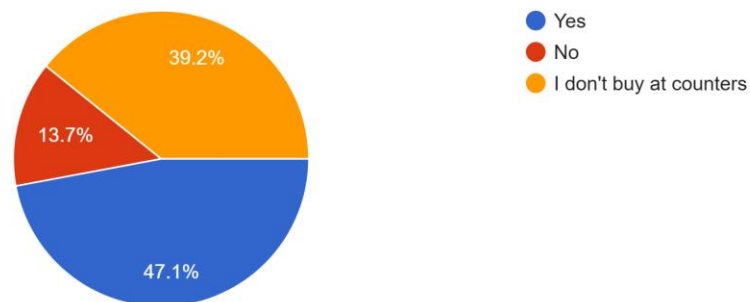
51 responses



Question 3:

Have you experienced long waiting times at bus ticket counters?

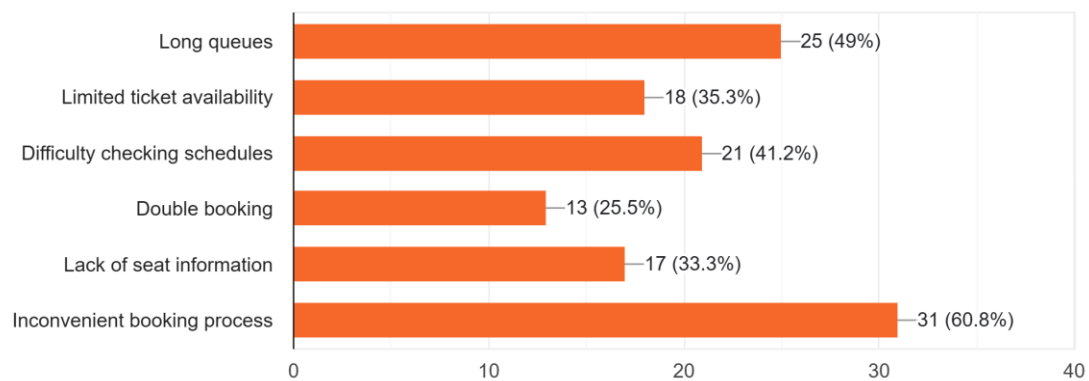
51 responses



Question 4:

Which of these problems have you faced when booking bus tickets?

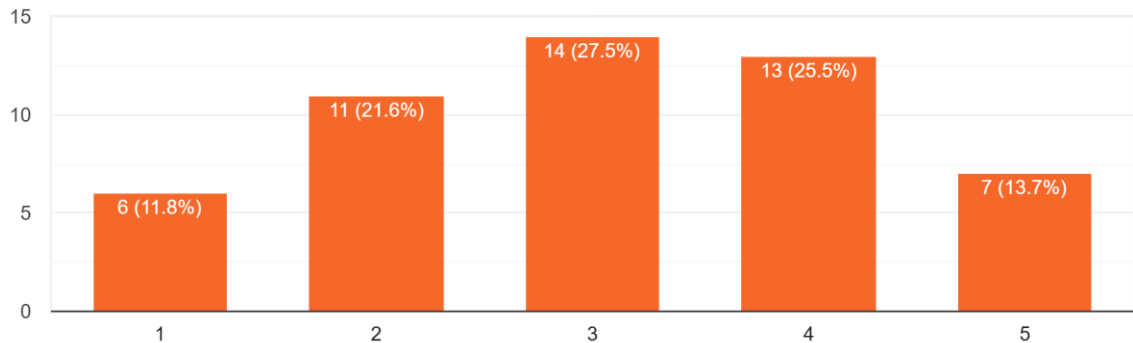
51 responses



Question 5:

How satisfied are you with the current bus ticket booking method?

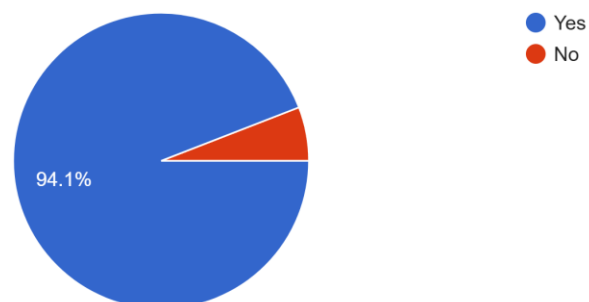
51 responses



Question 6:

Would you prefer an online bus ticket booking system?

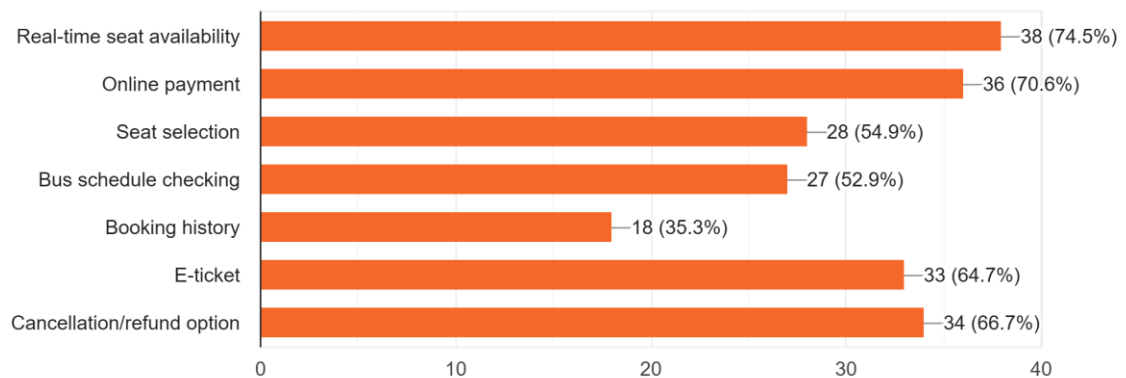
51 responses



Question 7:

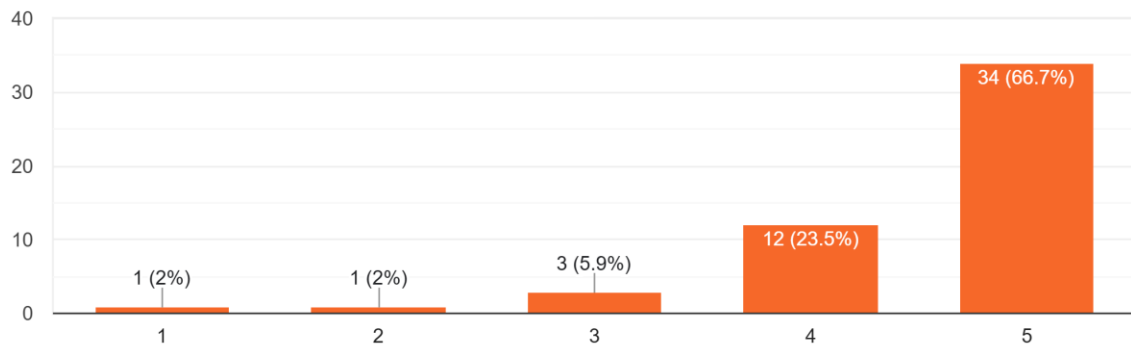
Which of these features would you like in bus ticket booking system

51 responses



Would an online booking system make your travel planning easier?

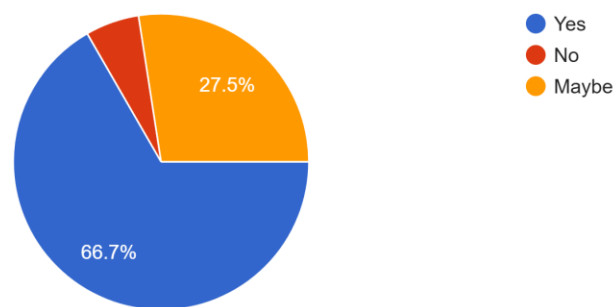
51 responses



Question 9:

Do you think an online booking system can reduce human errors?

51 responses



From analysing the questionnaire survey data, we observe a widespread demographic between those who use bus transportation very often and those who have never used it, with the largest percentage of respondents, 29.4%, using bus transportation frequently in a month.

We also observe that many respondents usually choose to buy their bus tickets remotely via online website or mobile application. We can cite that the choice may have stemmed from the fact that more than half of the respondents who have had prior experience buying tickets at counters have experienced long waiting times at said counters. Additionally, long queues are the second most common issue faced when buying tickets.

In order from most to least commonly faced problem when booking bus tickets, it is as such:

- Inconvenient booking process (60.8%)
- Long queues (49%)

- Difficulty checking schedules (41.2%)
- Limited ticket availability (35.3%)
- Lack of seat information (33.3%)
- Double booking (25.5%)

This culmination of different issues has led to mixed opinions in terms of satisfaction with the current state of bus ticket booking platforms, whether it be online or in-person.

Thus, we proceed to the final section of the survey, Section C, which is intended to determine the feasibility of the project. A whopping 94.1% of respondents agreed that they preferred an online bus ticket booking system. Additionally, many wished for a variety of different features in the platform, some of which proved more popular than the others.

In order from most to least wanted feature in an online bus ticket booking system, it is as such:

- Real-time seat availability (74.5%)
- Online payment (70.6%)
- Cancellation/refund option (66.7%)
- E-ticket (64.7%)
- Seat selection (54.9%)
- Bus schedule checking (52.9%)
- Booking history (35.3%)

In the beginning stages of developing and implementing a system, not all features can be included. As such, the survey serves as a method of identifying the most important features of a bus ticket booking system.

Majority of respondents agree that having an online booking system eases their travel planning easier. With this, one of the main goals of the project is achieved.

Furthermore, 66.7% of respondents believe that an online booking system can reduce human errors while 27.5% remain uncertain and 5.9% disbelieve it. For example, although an online booking system is still prone to technical errors, logistical errors such as double booking and long queues are easily solved by applying computer algorithms to smooth the booking process, thus preventing long queues from forming.

In conclusion, the final product of this project will address all major pain points that have been identified via the information gathered and package the solution into an easy-to-use final iteration.

5.0 Requirement Analysis

This section presents the requirement analysis based on the current AS-IS (manual) bus ticket booking system. The analysis is derived from the information gathered through surveys and interviews conducted with stakeholders and customers. It covers the current business

process, functional requirements, non-functional requirements, and the Logical Data Flow Diagrams (DFD) representing the existing system.

5.1 Current Business Process

5.1.1 Overview of Current System

The current bus ticket booking system operated by the company is entirely manual. Customers are required to physically visit the ticket counter at the bus terminal to purchase tickets. Counter staff manage all bookings manually using paper forms, printed schedule charts, and handwritten seat maps. There is no online platform, no real-time seat tracking, and no digital payment system in place.

5.1.2 Stakeholder Scenarios

The following scenarios describe how the current system is used by key stakeholders:

Scenario 1: Customer Books a Ticket

A customer who wishes to travel from Kuala Lumpur to Johor Bahru must travel to the bus terminal. After queuing at the counter, the customer verbally provides travel details (route, date, time) to the staff. The staff checks the printed schedule and manually verifies seat availability on a paper chart. If seats are available, the staff fills out a paper booking form with the customer's details, collects cash payment, calculates change manually, and writes a handwritten ticket. The entire process can take 15-30 minutes, especially during peak periods.

Scenario 2: Counter Staff Manages Bookings

Counter staff must simultaneously handle walk-in customers, manage multiple paper booking forms, and update seat charts manually. During busy periods (school holidays, public holidays), staff face high pressure, leading to data entry errors such as incorrect passenger names, wrong seat numbers, or double-booking of the same seat. At end-of-day, staff must manually tally all bookings and prepare a summary report for the admin.

Scenario 3: Admin Reviews Booking Records

The admin must physically collect paper booking forms from each counter to compile daily reports. This process is time-consuming and error prone as some forms may be misplaced or illegible. There is no centralised view of overall seat occupancy, revenue, or booking trends. Any schedule changes must be manually communicated to all counter staff.

5.1.3 Current Business Workflow (AS-IS)

The table below illustrates the step-by-step workflow of the current manual bus ticket booking process, including the actors involved and identified issues:

Step	Activity	Actor	Notes / Issues (AS-IS)
1	Customer travels to bus terminal ticket counter	Customer	Time-consuming; no remote access
2	Customer queues to see counter staff	Customer	Long queues during peak hours and holidays
3	Staff asks for travel details: route, date, number of passengers	Counter Staff	Manual verbal communication; error-prone
4	Staff manually checks physical schedule book or whiteboard for available buses	Counter Staff	No real-time update; schedules may be outdated
5	Staff manually checks available seats on printed seat chart	Counter Staff	Risk of double booking due to simultaneous counter sales
6	Staff records passenger name, IC number, and contact on paper booking form	Counter Staff	Handwriting errors, illegible records, risk of lost forms
7	Customer pays in cash; staff calculates total fare manually	Customer / Counter Staff	No digital payment; calculation errors possible
8	Staff manually writes out ticket and hands to customer	Counter Staff	No digital record; ticket can be lost or forged
9	Staff updates seat chart manually and files paper booking form	Counter Staff	No centralised database; admin retrieval difficult
10	At end of day, staff manually tallies bookings and reports to admin	Counter Staff / Admin	Time-consuming; errors in daily reports are common

Key points identified from this workflow align directly with the problems stated by stakeholders during interviews and questionnaire analysis:

- Customers must physically attend counter - no remote or online access
- No real-time seat availability - high risk of double-booking
- Manual record-keeping leads to data entry errors, lost records, and illegible forms



5.2 Functional Requirement (Input, Process and Output)

Functional requirements define what the Bus Ticket Booking System must do - describing the specific inputs, processes, and outputs required by users. Based on the AS-IS analysis and stakeholder feedback gathered in Section 4, the following functional requirements have been identified:

Functional requirements describe the functionality or system services and depend on the type of software, expected users, and the type of system where the software is used. They may be high-level statements (user requirements) or detailed descriptions of system services (system requirements).

Module	Input	Process	Output
Customer Registration	Full name, email address, phone number, password, IC/passport number	Validate and create customer account	Account created confirmation, unique customer ID generated
Customer Login & Authentication	Registered email and password	Verify credentials against customer database	Access granted / error message for invalid credentials
Schedule Search	Origin, destination, travel date, number of passengers	Search schedule database, filter available trips	List of available bus trips with time, price, and seat availability
Seat Selection	Selected trip ID, preferred seat number	Check seat availability in real-time, reserve seat temporarily	Seat map displayed, selected seat marked as reserved
Ticket Booking	Passenger details, selected seat, trip details	Validate booking details, record booking in database	Booking ID generated, booking summary displayed
Payment Processing	Payment method (card/e-wallet/FPX), booking ID, payment amount	Transmit payment to payment gateway, validate transaction	Payment receipt, transaction ID, booking status updated to Confirmed
E-Ticket Generation	Confirmed booking ID, passenger details, trip info	Generate e-ticket with QR code, send to customer email/SMS	E-ticket (PDF/digital) with QR code sent to customer

Booking History	Customer ID, date range (optional)	Retrieve all bookings for the customer from database	List of past and upcoming bookings with status
Booking Cancellation	Booking ID, cancellation reason	Validate cancellation policy, update booking status, initiate refund if applicable	Cancellation confirmation, refund notification (if eligible)
Admin - Manage Bus & Routes	Bus details (no., capacity, type), route info, schedule, fare	Add, edit, or delete bus/route/schedule records	Updated bus/route/schedule database, confirmation message
Admin - View Booking Reports	Report type, date range, route filter (optional)	Aggregate and filter booking/revenue data from database	Booking summary report, total revenue, occupancy rate

5.3 Non-Functional Requirement (Performance and Control)

Non-functional requirements (NFR) define the quality attributes, system properties, and constraints of the Bus Ticket Booking System. NFRs may be more critical than functional requirements - if these are not met, the system may be considered useless even if all functional requirements are fulfilled.

NFRs for this system are categorised into two groups: Performance Requirements and Control/Security Requirements.

5.3.1 Performance Requirements

Performance requirements address the speed, availability, and efficiency of the system:

No.	Requirement	Description	Metric / Measurement	Rationale
P1	Response Time	The system shall respond to user interactions (search, booking, login) within an acceptable time	Page load: < 3 seconds; Search results: < 2 seconds; Booking submission: < 5 seconds	Ensures a smooth user experience, especially during peak booking periods
P2	System Availability	The system shall be accessible to users at all times with minimal downtime	System uptime: $\geq 99.5\%$ per month; Planned maintenance windows: off-peak hours only	Bus booking often happens at any hour; customers must be able to book 24/7
P3	Concurrent Users	The system shall support multiple users performing bookings simultaneously without degradation	Minimum 200 concurrent users supported without performance drop	During peak seasons (holidays, long weekends), many users book at the same time
P4	Seat Availability Update	Seat availability must be updated in real-time to prevent double bookings	Seat status updated within 1 second of any booking or cancellation action	Prevents overbooking - a critical issue in the current manual system
P5	Data Processing Speed	Payment processing and booking confirmation shall be completed efficiently	Payment gateway response: < 10 seconds; Booking confirmation email/SMS sent within 30 seconds	Customers expect instant confirmation to reduce uncertainty after payment

5.3.2 Control Requirements (Security, Reliability and Usability)

Control requirements address the security, data integrity, reliability, and usability of the system:

No.	Requirement	Description	Metric / Measurement	Rationale
C1	User Authentication & Access Control	The system shall restrict access based on user roles (Customer, Admin)	Role-based login enforced; Admin functions inaccessible to customers; Session expiry after 30 minutes of inactivity	Prevents unauthorized manipulation of schedule, pricing, or booking records
C2	Data Security & Encryption	All sensitive data (personal info, payment details) shall be transmitted and stored securely	HTTPS/TLS 1.2+ for all communications; Passwords hashed using bcrypt; Payment data compliant with PCI-DSS standards	Protects customer personal data and financial information from breaches
C3	Input Validation & Error Handling	The system shall validate all user inputs and handle errors gracefully	All form fields validated before submission; Error messages displayed clearly without exposing system details; No SQL injection or XSS vulnerabilities	Prevents incorrect data entry (human errors) which was a key problem in the current system
C4	Data Backup & Recovery	Booking and payment data shall be backed up regularly to prevent data loss	Automated daily database backup; Recovery Point Objective (RPO): <= 24 hours; Recovery Time Objective (RTO): <= 4 hours	Eliminates the risk of lost records - a key weakness of the current manual system
C5	Audit Trail / Activity Logging	The system shall maintain logs of critical actions for accountability and troubleshooting	Logs stored for minimum 6 months; Logs capture: login attempts, booking creation/cancellation,	Enables traceability and accountability, especially important for payment disputes

			payment events, admin changes	and booking conflicts
C6	Usability	The system shall be easy to use by all users, including those with limited technical skills	New users able to complete a booking within 5 minutes without guidance; UI consistent across all pages; Mobile-responsive design supported	The wide demographic of bus users includes non-tech-savvy individuals; usability directly impacts adoption
C7	Portability & Compatibility	The system shall function correctly across major browsers and device types	Compatible with: Chrome, Firefox, Edge, Safari (latest 2 versions); Functional on desktop, tablet, and mobile devices	Supports cross-platform access requirement stated in project scope

5.4 Logical DFD AS-IS System

The Logical Data Flow Diagrams (DFD) below represent the current AS-IS bus ticket booking process. The DFDs model the flow of data through the existing system, identifying external entities, processes, data flows, and data stores. These diagrams are based on the Yourdon-Coad DFD notation.

Note: The DFDs are presented in tabular format below, describing the elements at each level. These should be drawn as formal DFD diagrams using standard notation (rectangles for external entities, rounded rectangles/circles for processes, open-ended rectangles for data stores, and arrows for data flows) in the final submission.

5.4.1 Context Diagram (Level 0 DFD)

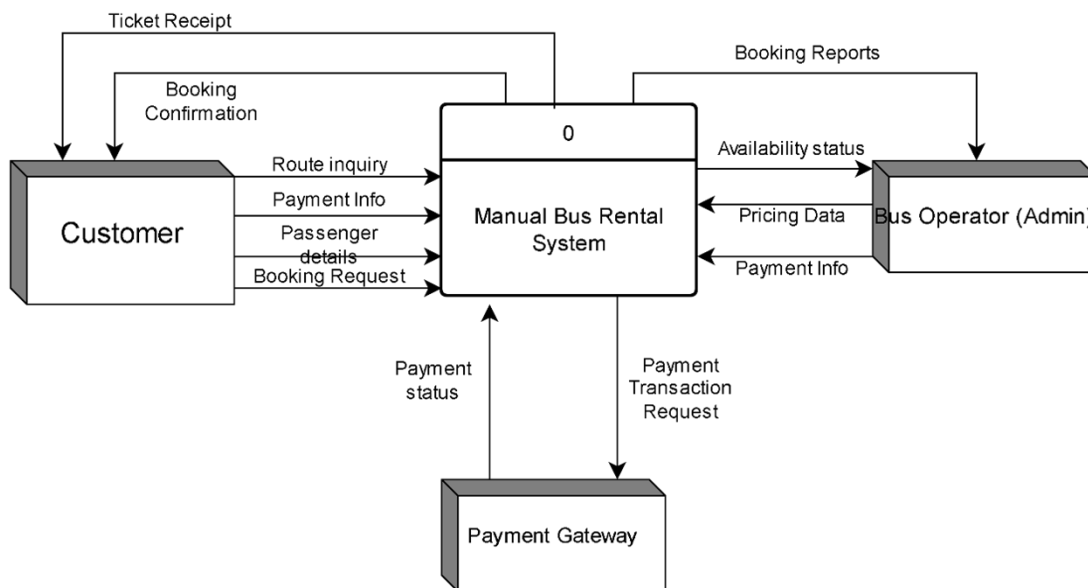
The Context Diagram provides the highest-level view of the system. It shows the main process (the entire Bus Ticket Booking System as a single process), the external entities that interact with it, and the data flows between them.

System: Bus Ticket Booking System | Process 0

External Entities: Customer, Bus Operator / Admin, Payment Gateway

Element	Name / Label	Description
External Entity	Customer	Books tickets, views schedules, makes payments
External Entity	Bus Operator (Admin)	Manages buses, routes, schedules, and booking data
External Entity	Payment Gateway	Processes payment transactions
Main Process	0 - Bus Ticket Booking System	The entire AS-IS manual booking system
Data Flow (in)	Booking request, passenger details, payment info, route enquiry	Customer inputs to system
Data Flow (out)	Booking confirmation, ticket receipt, schedule info, payment status	System outputs to customer
Data Flow (Admin in)	Route/schedule updates, bus info, pricing data	Admin inputs to system
Data Flow (Admin out)	Booking reports, revenue summary, seat availability status	System outputs to admin

Context Diagram - Data Flow Summary:



5.4.2 Diagram 0 (Level 1 DFD)

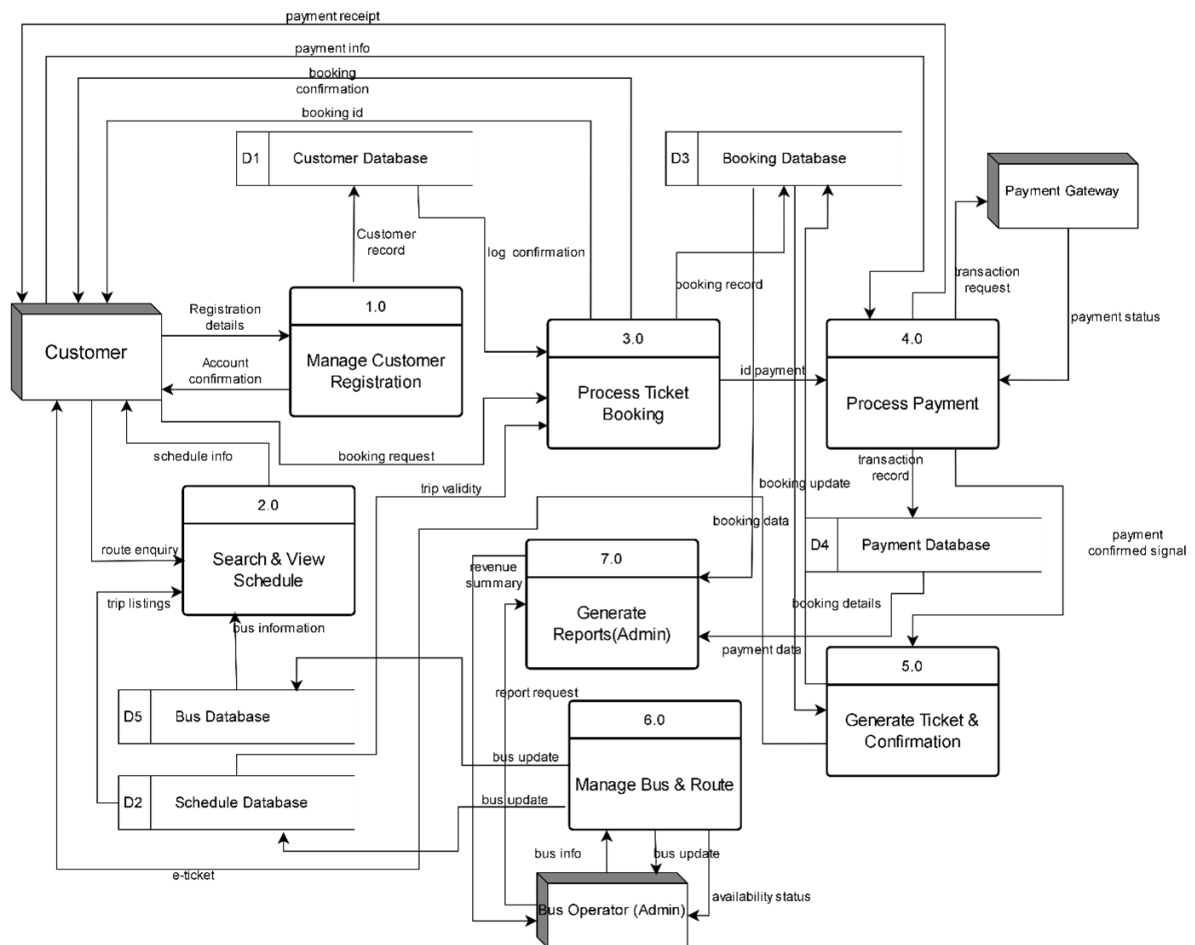
Diagram 0 expands Process 0 from the Context Diagram into the major sub-processes of the Bus Ticket Booking System. Each process represents a significant functional area of the system. Data stores represent databases or files used to store and retrieve information.

Data Stores identified in this system:

- D1 - Customer Database: Stores registered customer profiles, login credentials, and contact information
- D2 - Schedule Database: Stores bus routes, trip schedules, departure times, and fare information
- D3 - Booking Database: Stores all booking records, seat assignments, booking status, and booking history
- D4 - Payment Database: Stores transaction records, payment methods, and financial data
- D5 - Bus Database: Stores bus registration details, capacity, bus type, and operational status

Process No.	Process Name	Input Data Flow	Output Data Flow	Data Store
1.0	Manage Customer Registration	Customer details, login credentials	Account confirmation, login token	D1 - Customer DB
2.0	Search & View Schedule	Route enquiry, travel date	Schedule info, available seats	D2 - Schedule DB
3.0	Process Ticket Booking	Booking request, seat selection, passenger info	Booking confirmation, booking ID	D3 - Booking DB
4.0	Process Payment	Payment info (card/e-wallet), booking ID	Payment receipt, payment status	D4 - Payment DB
5.0	Generate Ticket & Confirmation	Confirmed booking ID, payment confirmation	E-ticket, confirmation email/SMS	D3 - Booking DB

6.0	Manage Bus & Route (Admin)	Bus info, route data, pricing, schedule	Updated schedule, confirmation	D2 - Schedule DB, D5 - Bus DB
7.0	Generate Reports (Admin)	Report criteria, date range	Booking reports, revenue summary	D3, D4 - Booking & Payment DB

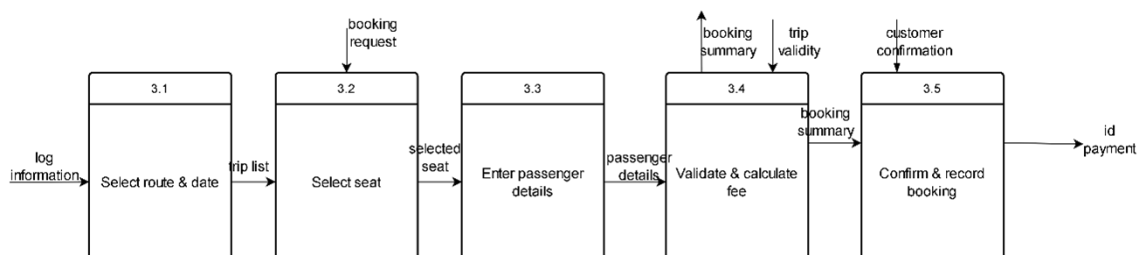


5.4.3 Child Diagram (Level 2 DFD) - Process 3.0: Process Ticket Booking

The child diagram further decomposes Process 3.0 (Process Ticket Booking), which is the core function of the system. This process was selected for decomposition as it involves the most complex data flows and is critical to the system's main objective of enabling online ticket booking.

Sub-Process	Process Name	Input Data Flow	Output Data Flow	Data Store
3.1	Select Route & Date	Origin, destination, travel date	Available trips list	D2 - Schedule DB
3.2	Select Seat	Trip ID, seat preference	Seat map, selected seat confirmation	D3 - Booking DB
3.3	Enter Passenger Details	Passenger name, IC/passport, contact number	Passenger details validated	D1 - Customer DB
3.4	Validate Booking	Passenger details, seat selection, trip info	Booking summary, total fare	D3 - Booking DB
3.5	Confirm & Record Booking	Validated booking data, payment confirmed	Booking ID, confirmation notice	D3 - Booking DB

Process 3.0 interacts with the following external entities and data stores:



6.0 Summary of Requirement Analysis Process

The requirement analysis process was conducted to identify the needs and expectations of potential users for the proposed Bus Ticket Booking System. Information was gathered through interviews and questionnaire surveys to obtain both detailed feedback and broader user opinions regarding the current bus ticket booking process.

The analysis of the current business process revealed several issues, including long waiting times at ticket counters, limited access to real-time information, and the possibility of human errors in managing booking records. The stakeholder scenarios and current workflow analysis further highlighted the need for a more efficient and systematic booking process.

Based on the findings, the functional requirements of the system were identified, including user registration, viewing bus schedules, checking seat availability, selecting seats, making bookings, and managing booking records. In addition, several non-functional requirements were established to ensure that the system operates efficiently, securely, and reliably while providing a user-friendly experience.

Overall, the requirement analysis confirms that the proposed Bus Ticket Booking System can address the weakness of the current system and meet the needs of both customers and bus operators. The identified requirements will serve as the foundation for the subsequent system design and development phases.

7.0 Appendices

Link for GitHub : [iqbal-faizal/Group1_Project1_SAD_20252026](https://github.com/iqbal-faizal/Group1_Project1_SAD_20252026): Project Management for
[System Analysis and Design](#)